

Haokai Ding

UNDERGRADUATE RESEARCHER · ROBOTICS

Shenzhen, China

✉ ditang0125@gmail.com | 🏠 haokaiding.github.io | 📷 HaokaiDing | ORCID: 0009-0001-6158-9897

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Abu Dhabi, UAE

M.Sc. IN ROBOTICS (INCOMING)

Fall 2026

Tsinghua University | Shenzhen X-Institute

Shenzhen, China

TSIEN EXCELLENCE IN ENGINEERING PROGRAM – X SCHOLAR

Sep. 2023 – Present

- Joint training program in underactuated robotics under the Tsien Excellence in Engineering Program (TEEP) of Tsinghua University.

Shenzhen Technology University, Future Technology School

Shenzhen, China

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY

Sep. 2022 – Jul. 2026

- Research focus: robotic systems, underactuated mechanisms, and LIBS spectroscopy.

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University

Shanghai, China

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION

Feb. 2026 – Present

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Shenzhen X-Institute, Tsinghua University

Shenzhen, China

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS

2023 – Present

- Designed and tested underactuated robotic grippers featuring semi-Peaucellier linkage and idle-stroke / delay-triggering mechanisms.
- Validated multiple prototypes through simulation and physical experiments; led several first-author submissions to IEEE robotics conferences.

Shenzhen Technology University

Shenzhen, China

UNDERGRADUATE RESEARCHER – LIBS SPECTROSCOPY FOR HEAVY METAL ANALYSIS

2022 – 2023

- Built and optimized a Laser-Induced Breakdown Spectroscopy (LIBS) system for rapid heavy-metal adsorption analysis.
- Improved the calibration pipeline and acquisition accuracy; contributed to a peer-reviewed publication.

Publications

Legend: [†] Co-first author; * Corresponding author. Listed in reverse chronological order by IEEE Xplore date added.

SELECTED CONFERENCE PAPERS (ROBOTICS)

Y. Chen, H. Luan, H. Ding, W. Zhang*

IEEE RAAI 2025, Singapore

[P1] AN IDLE-STROKE UNDERACTUATED GRIPPER WITH INDEPENDENT DOUBLE-PARALLELOGRAM LINKAGES FOR PINCHING AND ADAPTIVE GRASPING

Mar. 2026

H. Ding[†], X. Zhong, W. Huang, T. Yang*, W. Zhang*

IEEE ROBIO 2025, Bangkok, Thailand

[P2] BIOCREST – A FLEXIBLE ADAPTIVE ROBOTIC HAND BASED ON IDLE-STROKE MECHANISM

Feb. 2026

H. Ding, H. Luan, T. Yang*, W. Zhang*

IEEE ICCMA 2025, Macau, China

[P3] GLINT: AN IDLE-STROKE GRASP-AND-LIFT HAND FOR IN-HAND MANIPULATION

Feb. 2026

H. Ding[†], J. Fan, Z. Zhu, Y. Zhao, K. Chen*, W. Zhang*

IEEE ICARM 2025, Hangzhou, China

[P4] PEAUCELLIER GRIPPER: A NOVEL UNDERACTUATED GRIPPER FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASPING

Dec. 2025

H. Ding[†], Y. Chen[†], D. Jia, W. Zhang*

IEEE ICIA 2025, Wuhan, China

[P5] AN UNDERACTUATED GRIPPER WITH GRASSHOPPER-INSPIRED LINKAGES AND DELAY TRIGGERING FOR PINCHING AND ADAPTIVE GRASPING

Dec. 2025

H. Ding[†], W. Zhang*

IEEE/RSJ IROS 2025, Hangzhou, China

[P6] A NOVEL GRIPPER WITH SEMI-PEAUCELLIER LINKAGE AND IDLE-STROKE MECHANISM FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASPING

Nov. 2025

OTHER PUBLICATIONS

Y. Liu[†], H. Ding[†], H. Chen, H. Gao, J. Yu, F. Mo, N. Wang^{*}

[P8] RECENT PROGRESS ON THE RESEARCH OF 3D PRINTING IN AQUEOUS ZINC-ION BATTERIES

Polymers (MDPI), 17(15):2136

Aug. 2025

H. Xie, L. You, X. Fang, L. Li, H. Ding, Z. Zhou, G. Zhang, D. Zhang

E3S Web of Conferences, 615 (2025)

02010

[P9] RAPID ANALYSIS OF HEAVY METAL ELEMENT ADSORPTION BY SCG BASED ON LIBS TECHNOLOGY

Feb. 2025

Honors & Awards

INTERNATIONAL AWARDS

2025 Conference Session Chair, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Hangzhou, China

DOMESTIC AWARDS

2025 National Scholarship, Ministry of Education of China

China

2025 Undergraduate Champion, ASME Student Mechanism & Robot Design Competition

China

2023 Gold Prize, 9th China "Internet+" Innovation Competition (Guangdong Division)

Guangdong, China

2023 Second Prize, 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division)

Guangdong, China

2024 President's Award, Shenzhen Technology University

Shenzhen, China

2024 Research and Innovation Award (First Prize), Shenzhen Technology University

Shenzhen, China

2023 Research and Innovation Award (First Prize), Shenzhen Technology University

Shenzhen, China

Service

- Conference Reviewer – IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025.
- Conference Reviewer – IEEE International Conference on Automation Science and Engineering (CASE).
- Session Chair – IEEE/RSJ IROS 2025.